

# Category Theory and Functional Programming

## Some Basic Constructions

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# Preliminaries

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## Convention

The number assigned to chapters, examples, exercises, figures, pages, sections, and theorems on these slides correspond to the numbers assigned in the textbook (Abramsky and Tzevelekos 2011).

# Outline

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Initial and Terminal Objects

Products

Coproducts

References

# Initial and Terminal Objects

# Initial and Terminal Objects

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## Introduction

We shall introduce abstract characterisations of the empty set and the one-element sets in set theory.

# Initial and Terminal Objects

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## Definition

Let  $\mathcal{C}$  be a category. An object  $0$  in  $\mathcal{C}$  is **initial** iff for any object  $A$  there is a **unique** arrow (universal property)

$$0 \rightarrow A.$$

# Initial and Terminal Objects

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## Definition

Let  $\mathcal{C}$  be a category. An object  $0$  in  $\mathcal{C}$  is **initial** iff for any object  $A$  there is a **unique** arrow (universal property)

$$0 \rightarrow A.$$

## Definition

Let  $\mathcal{C}$  be a category. An object  $1$  in  $\mathcal{C}$  is **terminal** iff for any object  $A$  there is a **unique** arrow (universal property)

$$A \rightarrow 1.$$

# Initial and Terminal Objects

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## Observation

Initial and terminal objects are **dual** notions.



# Initial and Terminal Objects

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## Example

- In **Set**, the **empty set** is an initial object and any **one-element set** is a terminal object.
- In **Pos**, the poset  $(\emptyset, \emptyset)$  is an initial object and the poset  $(\{*\}, \{(*, *)\})$  is a terminal object.
- In **Top**, the topological space  $(\emptyset, \{\emptyset\})$  is an initial object and the topological space  $(\{*\}, \{\emptyset, \{*\}\})$  is a terminal object.

# Initial and Terminal Objects

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## Example

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- In **Pos**, the poset  $(\emptyset, \emptyset)$  is an initial object and the poset  $(\{*\}, \{(*, *)\})$  is a terminal object.
- In **Top**, the topological space  $(\emptyset, \{\emptyset\})$  is an initial object and the topological space  $(\{*\}, \{\emptyset, \{*\}\})$  is a terminal object.

## Exercise

Verify the initial and terminal objects in the previous example. In each case, identify the canonical arrows (Exercise 18).

# Initial and Terminal Objects

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## Exercise

For the category **Rel**, identify the initial and terminal objects, and the canonical arrows (Exercise 19).

## Exercise

Suppose that a monoid, viewed as a category, has either an initial or a terminal object. What must the monoid be? (Exercise 20).

# Initial and Terminal Objects

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## Example

In a poset, seen as a category,

- (i) an object is initial iff it is the least element,
- (ii) an object is terminal iff it is the greatest element.

# Initial and Terminal Objects

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## Example

In a poset, seen as a category,

- (i) an object is initial iff it is the least element,
- (ii) an object is terminal iff it is the greatest element.

## Question

Does a category need to have either an initial object or a terminal object?

# Initial and Terminal Objects

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## Example

In a poset, seen as a category,

- (i) an object is initial iff it is the least element,
- (ii) an object is terminal iff it is the greatest element.

## Question

Does a category need to have either an initial object or a terminal object?

*Answer:* No. The poset  $(\mathbb{Z}, \leq)$ , seen as a category, has neither.

# Initial and Terminal Objects

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## Example

For **Hask**, the `Void` data type<sup>†</sup> is an initial object.

```
data Void

absurd :: Void -> a
absurd a = case a of {}
```

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<sup>†</sup>From the module `Data.Void` of the base library.

# Initial and Terminal Objects

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## Example

For **Hask**, the `Unit` data type is a terminal object.

```
data Unit = MkUnit

t :: a -> Unit
t _ = MkUnit
```



# Initial and Terminal Objects

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## Example

For **Hask**, the `Unit` data type is a terminal object.

```
data Unit = MkUnit

t :: a -> Unit
t _ = MkUnit
```

The terminal object is built-in as `()` whose unique term is `()`, that is, `() :: ()`.

# Initial and Terminal Objects

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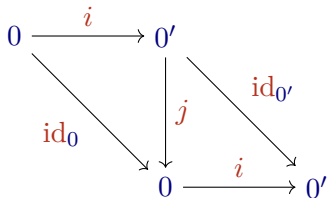
## Theorem (Proposition 21)

Initial objects are unique up to isomorphism, that is, if  $0$  and  $0'$  are initial objects in a category  $\mathcal{C}$  then there exists a unique isomorphism  $0 \xrightarrow{\cong} 0'$ .

# Initial and Terminal Objects

Proof.

Let  $0$  and  $0'$  be initial objects in a category  $\mathcal{C}$ . Because  $0$  and  $0'$  are initial objects we have that the following diagram commutes:



$$\begin{pmatrix} j \circ i = \text{id}_0 \\ i \circ j = \text{id}_{0'} \end{pmatrix}$$

That is, there is a unique isomorphism  $i : 0 \xrightarrow{\cong} 0'$ .



# Initial and Terminal Objects

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## Theorem

Terminal objects are unique up to isomorphism.

## Exercise

Prove the previous theorem.

Products

# Products

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## Introduction

We shall introduce abstract characterisations of products (e.g. Cartesian products of sets and direct products of groups).

# Binary Products

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## Example (Cartesian product in set theory)

(i) Let  $X$  and  $Y$  be sets. The **Cartesian product** of  $X$  and  $Y$  is defined by

$$X \times Y := \{ (x, y) \mid x \in X \wedge y \in Y \},$$

where the **ordered pair**  $(x, y)$  can be defined by

$$(x, y) := \{ \{x, y\}, y \} \quad (\text{Kuratowski's definition})$$

and it satisfies that

$$(x, y) = (x', y') \quad \text{iff} \quad x = x' \text{ and } y = y'.$$

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# Binary Products

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Example (Cartesian product in set theory (continuation))

(ii) Two **coordinate projections** on  $X \times Y$  are defined by

$$\pi_1 : X \times Y \rightarrow X := (x, y) \mapsto x,$$

$$\pi_2 : X \times Y \rightarrow Y := (x, y) \mapsto y,$$

where

$$c = (\pi_1 c, \pi_2 c), \quad \text{for all } c \in X \times Y.$$

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# Binary Products

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Example (Cartesian product in set theory (continuation))

(iii) Let  $f : Z \rightarrow X$  and  $g : Z \rightarrow Y$ . The **pair  $f$  and  $g$**  function is defined by

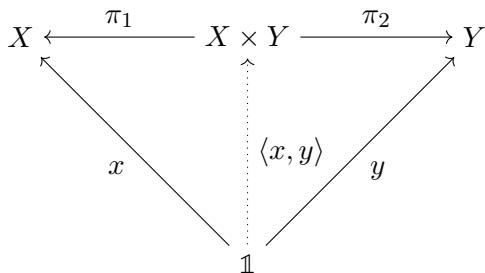
$$\langle f, g \rangle : Z \rightarrow X \times Y := z \mapsto (f z, g z).$$

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# Binary Products

## Example (Cartesian product in set theory (continuation))

(iv) We state the Cartesian product properties by saying that the following diagram commutes.



$$\begin{pmatrix} \pi_1 \circ \langle x, y \rangle = x \\ \pi_2 \circ \langle x, y \rangle = y \end{pmatrix}$$

# Binary Products

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## Definition

Let  $A_1$  and  $A_2$  be objects in a category  $\mathcal{C}$ . A **binary product** of  $A_1$  and  $A_2$  is a triple  $(P, \pi_1, \pi_2)$ , where  $P$  is an object in  $\mathcal{C}$ , denoted  $A_1 \times A_2$ , and  $\pi_1$  and  $\pi_2$  are two arrows

$$A_1 \xleftarrow{\pi_1} A_1 \times A_2 \xrightarrow{\pi_2} A_2,$$

such that for every object  $B$  and arrows

$$A_1 \xleftarrow{f_1} B \xrightarrow{f_2} A_2$$

there exists an **unique** arrow

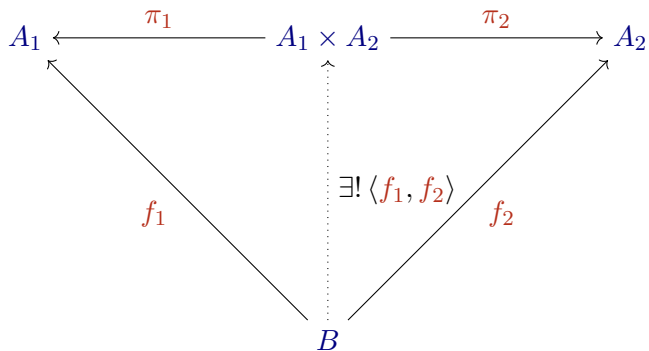
$$\langle f_1, f_2 \rangle : B \rightarrow A_1 \times A_2$$

such that the following diagram commutes (universal property):

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# Binary Products

## Definition (continuation)



$$\begin{pmatrix} \pi_1 \circ \langle f_1, f_2 \rangle = f_1 \\ \pi_2 \circ \langle f_1, f_2 \rangle = f_2 \end{pmatrix}$$

# Binary Products

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## Example

- In **Set**, products are the Cartesian products.
- In **Pos**, products are Cartesian products with the product order.<sup>†</sup>
- In **Top**, products are Cartesian products with the product topology.

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<sup>†</sup>The textbook uses “pointwise order” instead of “product order”.

# Binary Products

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## Example

- In **Set**, products are the Cartesian products.
- In **Pos**, products are Cartesian products with the product order.<sup>†</sup>
- In **Top**, products are Cartesian products with the product topology.

## Exercise

Verify the previous claims (Exercise 19).

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<sup>†</sup>The textbook uses “pointwise order” instead of “product order”.

# Binary Products

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## Definition

A category  $\mathcal{C}$  **has binary products** iff each pair of objects of  $\mathcal{C}$  have a binary product.

# Binary Products

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## Example

Since it is possible to define the Cartesian product between any pair of sets, the category **Set** has binary products.



# Binary Products

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## Example

Since it possible to define the Cartesian product between any pair of sets, the category **Set** has binary products.

## Example

In a poset, seen as a category, products are (binary) greatest lower bounds (meets). This category has not binary products.

# Binary Products

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## Exercise

Prove Proposition 27.

## Exercise

Prove Proposition 28.

# Ternary Products

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## Definition

Let  $A_1$ ,  $A_2$  and  $A_3$  be objects in a category  $\mathcal{C}$ . A **ternary product** of  $A_1$ ,  $A_2$  and  $A_3$  is a quadruple

$$(P, \pi_1, \pi_2, \pi_3),$$

where  $P$  is an object in  $\mathcal{C}$ , denoted  $A_1 \times A_2 \times A_3$ , and  $\pi_1, \pi_2, \pi_3$  are arrows from  $A_1 \times A_2 \times A_3$  to  $A_1, A_2, A_3$ , respectively, such that **for every** object  $B$  and arrows  $f_1, f_2, f_3$  from  $B$  to  $A_1, A_2, A_3$ , respectively, there exists an **unique** arrow

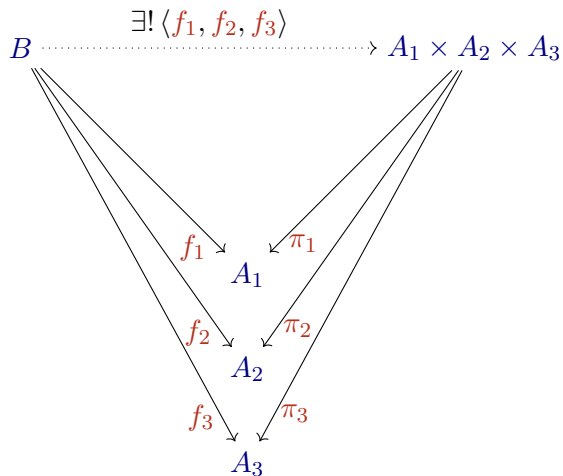
$$\langle f_1, f_2, f_3 \rangle : B \rightarrow A_1 \times A_2 \times A_3$$

such that the following diagram commutes (universal property):

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# Ternary Products

## Definition (continuation)



$$\begin{pmatrix} \pi_1 \circ \langle f_1, f_2, f_3 \rangle = f_1 \\ \pi_2 \circ \langle f_1, f_2, f_3 \rangle = f_2 \\ \pi_3 \circ \langle f_1, f_2, f_3 \rangle = f_3 \end{pmatrix}$$

# Nullary Products

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## Observation

By removing the objects  $A_i$  (which also remove the projections  $\pi_i$  and the equations  $\pi_i \circ \langle f_i \rangle = f_i$ ) from the binary (or ternary) products, we get the nullary products.

# Nullary Products

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## Observation

By removing the objects  $A_i$  (which also remove the projections  $\pi_i$  and the equations  $\pi_i \circ \langle f_i \rangle = f_i$ ) from the binary (or ternary) products, we get the nullary products.

## Definition

A **nullary product** in a category  $\mathcal{C}$  is an object  $P$ , such that **for any** object  $B$ , there is a **unique** arrow  $B \rightarrow P$  (universal property).

# Nullary Products

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## Observation

By removing the objects  $A_i$  (which also remove the projections  $\pi_i$  and the equations  $\pi_i \circ \langle f_i \rangle = f_i$ ) from the binary (or ternary) products, we get the nullary products.

## Definition

A **nullary product** in a category  $\mathcal{C}$  is an object  $P$ , such that for any object  $B$ , there is a unique arrow  $B \rightarrow P$  (universal property).

## Observation

Note that the above object  $P$  is just a terminal object of  $\mathcal{C}$ .

# Nullary Products

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## Exercise

What is the product of the empty family? (Exercise 29)



# Finite Products

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## Definition

A category **has finite products** iff the category has products for all  $n \in \mathbb{N}$ .

# Finite Products

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## Exercise

Show that if a category has binary and nullary products then it has finite products (Exercise 30).

# General Products

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## Introduction

We shall generalise finite products to products of arbitrary objects.

# General Products

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## Example (Cartesian product of a family of sets)

- (i) Let  $\{X_i\}_{i \in I}$  be a family of sets indexed by  $I$ . The **Cartesian product** of the family of sets  $\{X_i\}_{i \in I}$  is defined by

$$\prod_{i \in I} X_i = \left\{ f : I \rightarrow \bigcup_{i \in I} X_i \mid \text{for all } i \in I, f(i) \in X_i \right\}.$$

# General Products

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$$\prod_{i \in I} X_i = \left\{ f : I \rightarrow \bigcup_{i \in I} X_i \mid \text{for all } i \in I, f(i) \in X_i \right\}.$$

- (ii) For  $i \in I$ , the  **$i$ th-coordinate projection** map is defined by

$$\pi_i : \left( \prod_{j \in J} X_j \right) \rightarrow X_i := f \mapsto f(i).$$

# General Products

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## Definition

Let  $\{A_i\}_{i \in I}$  be a family of objects in a category  $\mathcal{C}$ . A product for the family  $\{A_i\}_{i \in I}$  is an object  $\prod_{i \in I} A_i$  and arrows

$$\pi_i : \left( \prod_{i \in I} A_i \right) \rightarrow A_i$$

such that for every object  $B$  and arrows

$$f_i : B \rightarrow A_i$$

there exists an unique arrow

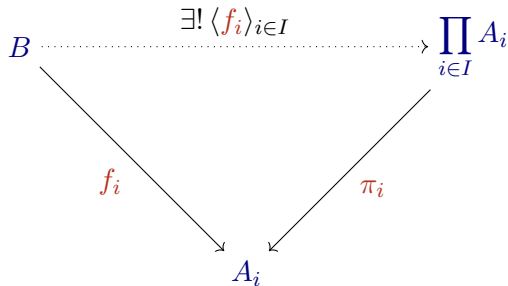
$$\langle f_i \rangle_{i \in I} : B \rightarrow \prod_{i \in I} A_i$$

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# General Products

## Definition (continuation)

such that, for  $i \in I$ , the following diagram commutes (universal property):



$$\left( \pi_i \circ \langle f_i \rangle_{i \in I} = f_i \right)$$

# Coproducts



# Coproducts

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## Introduction

We shall introduce abstract characterisations of disjoint unions (also called disjoint sums).

# Binary Coproducts

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## Example (Disjoint union in set theory)

(i) Let  $X$  and  $Y$  be sets. The **disjoint union** of  $X$  and  $Y$  is defined by

$$\begin{aligned} X + Y &:= (\{1\} \times X) \cup (\{2\} \times Y) \\ &= \{ (1, x) \mid x \in X \} \cup \{ (2, y) \mid y \in Y \}. \end{aligned}$$

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# Binary Coproducts

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Example (Disjoint union in set theory (continuation))

(ii) Two **injections** for  $X + Y$  are defined by

$$\text{in}_1 : X \rightarrow X + Y := x \mapsto (1, x),$$

$$\text{in}_2 : Y \rightarrow X + Y := y \mapsto (2, y).$$

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# Binary Coproducts

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## Example (Disjoint union in set theory (continuation))

(iii) Let  $f : X \rightarrow Z$  and  $g : Y \rightarrow Z$ . The **case**  $f$  **or**  $g$  function is defined by

$$[f, g] : X + Y \rightarrow Z$$

$$[f, g] (1, x) := f x,$$

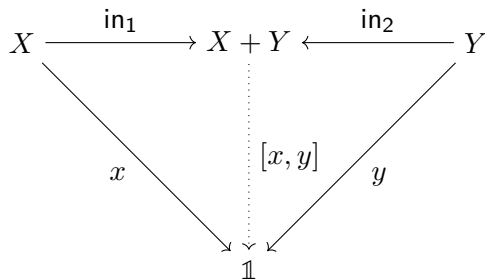
$$[f, g] (2, y) := g y.$$

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# Binary Coproducts

## Example (Disjoint union set theory (continuation))

(iv) We state the disjoint union properties by saying that the following diagram commutes.



$$\begin{pmatrix} [x, y] \circ \text{in}_1 = x \\ [x, y] \circ \text{in}_2 = y \end{pmatrix}$$

# Binary Coproducts

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## Definition

Let  $A_1$  and  $A_2$  be objects in a category  $\mathcal{C}$ . A **binary coproduct** of  $A_1$  and  $A_2$  is a triple  $(P, \text{in}_1, \text{in}_2)$ , where  $P$  is an object in  $\mathcal{C}$ , denoted  $A_1 + A_2$ , and  $\text{in}_1$  and  $\text{in}_2$  are two arrows

$$A_1 \xrightarrow{\text{in}_1} A_1 + A_2 \xleftarrow{\text{in}_2} A_2,$$

such that for every object  $B$  and arrows

$$A_1 \xrightarrow{f_1} B \xleftarrow{f_2} A_2$$

there exists an **unique** arrow

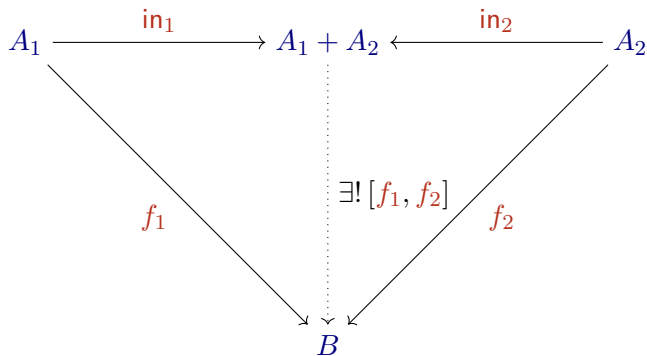
$$[f_1, f_2] : A_1 + A_2 \rightarrow B$$

such that the following diagram commutes (universal property):

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# Binary Coproducts

## Definition (continuation)



$$\begin{pmatrix} [f_1, f_2] \circ \text{in}_1 = f_1 \\ [f_1, f_2] \circ \text{in}_2 = f_2 \end{pmatrix}$$

# Binary Coproducts

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## Example

- In **Set**, disjoint unions are binary coproducts.
- In **Pos**, disjoint unions are binary coproducts.
- In **Top**, topological disjoint unions are binary coproducts.



# Binary Coproducts

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## Example

- In **Set**, disjoint unions are binary coproducts.
- In **Pos**, disjoint unions are binary coproducts.
- In **Top**, topological disjoint unions are binary coproducts.

## Exercise

Verify the previous claims (Exercise 33).

# Binary Coproducts

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## Example

In a poset, seen as a category, binary coproducts are (binary) least upper bounds (joins).

# Binary Coproducts

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## Example

In a poset, seen as a category, binary coproducts are (binary) least upper bounds (joins).

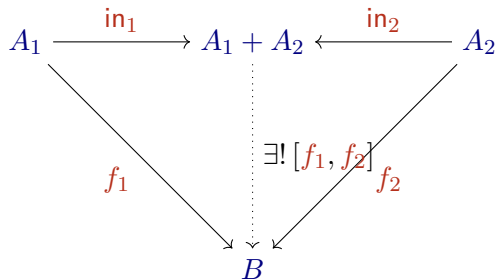
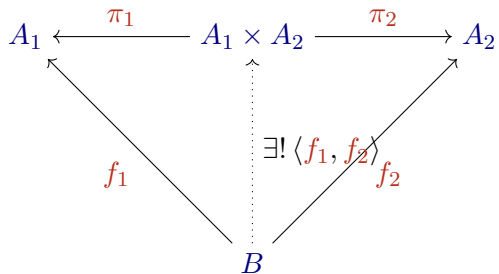
## Observation

The previous example show that, a difference of the disjoint union in set theory, the binary coproduct between any pair of objects of a category may not exist.

# Binary Coproducts

## Duality

Binary products and binary co-products are dual notions.



## References

## References

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S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: [10.1007/978-3-642-12821-9\\_1](https://doi.org/10.1007/978-3-642-12821-9_1) (cit. on p. 2).